

When Corruptions Compound: Superadditive Degradation of Vision-Language Encoders under Combined Image Corruptions

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Abstract

Existing robustness benchmarks for Vision-Language Models (VLMs), including ImageNet-C [1] and RobustBench [2], evaluate models under *individual* image corruptions applied in isolation. Real-world deployments, however, routinely impose *multiple simultaneous* corruptions—sensor noise co-occurring with fog, motion blur combined with low contrast. We present the first systematic study of *pairwise corruption interactions* for CLIP-family encoders under zero-shot classification. Using three OpenCLIP [3] models (ViT-B/32, ViT-B/16, ViT-L/14) and seven corruption types at three severity levels, we evaluate all $\binom{7}{2} = 21$ pairwise combinations and quantify *interaction effects* as the gap between observed and expected (additive) accuracy drops. Our key findings are: (i) **Superadditivity is real**—at low severity, up to 8 of 21 pairs degrade accuracy worse than the sum of individual drops; (ii) **contrast + shot_noise** is the most dangerous pair across all architectures (interaction up to +0.227); (iii) **model scale amplifies superadditivity**—ViT-L/14 shows twice the interaction effect of ViT-B/32; (iv) **floor effects suppress superadditivity** at high severity, revealing a regime transition absent from current benchmarks; (v) **corruption order is critical**—4 of 5 top pairs are order-sensitive, with accuracy gaps up to 0.21 between application orders, exposing a worst-case evaluation gap in existing protocols. These findings collectively expose systematic blind spots absent from all existing VLM robustness benchmarks, including ImageNet-C [1], CCC [4], and related work on augmentation–corruption interactions [5]. Code is available at <https://github.com/anonymous/vlm-combined-corruptions>.

1 Introduction

Vision-Language Models (VLMs) such as CLIP [6] have demonstrated strong zero-shot generalization across visual recognition benchmarks. Their robustness to distribution shift has been studied through ImageNet-C [1], which applies 15 corruption types at 5 severity levels, and RobustBench [2], which tracks state-of-the-art robustness over time.

The gap. A fundamental assumption of all existing corruption benchmarks is that corruptions are applied *one at a time*. This

assumption is violated in virtually every real-world deployment scenario. A camera on an autonomous vehicle simultaneously experiences sensor noise, motion blur, and reduced contrast from adverse weather. A surveillance system processes frames degraded by JPEG compression *and* brightness variation. fig. 1 illustrates these compound effects.

Research questions.

- RQ1.** Do pairwise corruption combinations degrade VLM accuracy *superadditively*—worse than the sum of individual drops?
- RQ2.** Does model scale (ViT-B/32 $\rightarrow$ ViT-B/16 $\rightarrow$ ViT-L/14) confer robustness against combined corruptions?
- RQ3.** Which pairs are most dangerous, and is the ranking consistent across architectures?
- RQ4.** Does the *order* of corruption application affect outcomes?

Contributions.

- We introduce the **interaction effect** metric $\mathcal{I}$ for quantifying superadditivity in corruption pairs (section 3).
- We conduct the **first systematic pairwise evaluation** of combined corruptions for CLIP-family encoders in zero-shot classification.
- We identify **contrast + shot_noise** as universally the most dangerous pair, with $\mathcal{I} = +0.227$ for ViT-L/14.
- We reveal a **severity-dependent regime transition**: superadditivity dominates at low severity and collapses at high severity.
- We demonstrate that **corruption order is critical**: 4 of 5 top pairs are order-sensitive, with worst-case accuracy gaps up to 0.21.
- We show a **patch-size trade-off**: smaller patches improve noise robustness at the cost of JPEG robustness.

2 Related Work

Corruption robustness benchmarks. Hendrycks and Dietterich [1] introduced ImageNet-C, applying 15 corruptions *independently* at 5 severity levels and aggregating results via mean Corruption Error (mCE). This single-corruption protocol is inherited by ImageNet-3DCC [7], RobustBench [2], and all subsequent corruption benchmarks.

VLM robustness. Recent work has studied CLIP under distributional shift. The ROZ benchmark [8] evaluates zero-shot CLIP under natural, synthetic, and adversarial shifts—all applied individually. BAT-CLIP [9] shows that zero-shot CLIP is not robust to common corruptions under increasing severity, again with independent corruptions. A concurrent survey [10] finds that robustness does not necessarily scale with model size, consistent with our results, but does not study combined corruptions.

Combined corruptions. The Continuously Changing Corruptions (CCC) dataset [4] combines pairs of corruptions via random walks to benchmark *continual test-time adaptation* (TTA) methods for CNNs. CCC differs fundamentally from our work: it targets temporal corruption streams for TTA evaluation on ResNet-50, and does not quantify pairwise interaction effects. **To the best of our knowledge, no prior work has systematically quantified pairwise corruption interaction effects for VLM encoders in zero-shot classification.**

3 Methodology

3.1 Models

We evaluate three OpenCLIP [3] models trained on LAION-2B [11]: **ViT-B/32** (patch size 32, 88M params), **ViT-B/16** (patch size 16, 86M params), and **ViT-L/14** (patch size 14, 307M params). All models use the prompt template "a photo of a {class}." for zero-shot classification.

3.2 Dataset

We use the CIFAR-100 [12] test split (10,000 images, 100 classes), sampling $N = 1,000$ images with fixed seed 42.

3.3 Corruption Protocol

We use the `imagecorruptions` library [13], which implements the ImageNet-C corruption suite. Seven corruption types are selected: *noise* (gaussian_noise, shot_noise), *blur* (motion_blur), *weather* (fog), and *digital* (jpeg_compression, brightness, contrast). Severity levels $s \in \{1, 3, 5\}$ are evaluated.

Table 1: Top-1 accuracy under clean, average single-corruption (7 types), and average combined-corruption (21 pairs) conditions at severity $s = 1$. **Gap** = Single – Combined.

Model	Clean	Single	Combined	Gap
ViT-B/32	0.753	0.450	0.218	0.231
ViT-B/16	0.788	0.473	0.239	0.234
ViT-L/14	0.835	0.544	0.295	0.249

Combined corruptions apply two corruptions sequentially:

$$\mathbf{x}_{c_1 \rightarrow c_2} = \text{corrupt}(\text{corrupt}(\mathbf{x}, c_1, s), c_2, s). \quad (1)$$

All $\binom{7}{2} = 21$ pairwise combinations are evaluated.

3.4 Interaction Effect Metric

Let $\text{acc}(\cdot)$ denote Top-1 accuracy. The **accuracy drop** under corruption c at severity s is:

$$\Delta(c, s) = \text{acc}_{\text{clean}} - \text{acc}(c, s). \quad (2)$$

Under the **additive independence assumption**, the expected drop under combined corruptions would be:

$$\Delta_{\text{exp}}(c_1, c_2, s) = \Delta(c_1, s) + \Delta(c_2, s). \quad (3)$$

The **interaction effect** is:

$$\mathcal{I}(c_1, c_2, s) = \Delta(c_1 + c_2, s) - \Delta_{\text{exp}}(c_1, c_2, s). \quad (4)$$

A pair is classified using threshold $\varepsilon = 0.005$:

$$\begin{aligned} \mathcal{I} > +\varepsilon &\Rightarrow \text{superadditive} \\ |\mathcal{I}| \leq \varepsilon &\Rightarrow \text{additive} \\ \mathcal{I} < -\varepsilon &\Rightarrow \text{subadditive} \end{aligned}$$

4 Experiments & Results

4.1 Experimental Setup

All experiments run on a single CUDA GPU with batch size 64. Text features are precomputed once per model. Total compute: ≈ 111 minutes.

4.2 RQ1: Is Superadditivity Real?

table 2 confirms that superadditivity is a robust phenomenon at low severity ($s = 1$): across all three models, 6–8 of 21 pairs exhibit $\mathcal{I} > 0.005$. table 3 shows the five most dangerous pairs.

contrast + shot_noise is the top-ranked superadditive pair for *all three* architectures, with interaction effects of +0.111, +0.118, and +0.227 for ViT-B/32, ViT-B/16, and ViT-L/14, respectively.

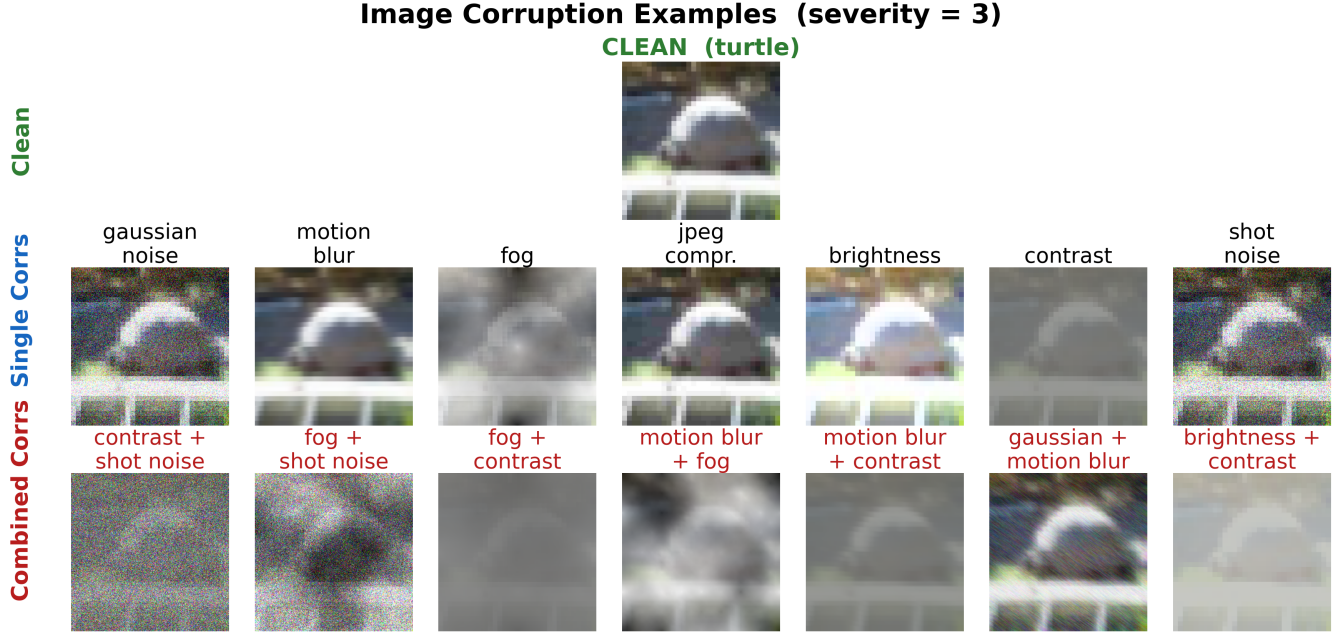


Figure 1: Example images from CIFAR-100 under individual and combined corruptions at severity $s = 3$. **Row 1:** clean. **Row 2:** single corruptions (left to right: gaussian_noise, motion_blur, fog, jpeg_compression, brightness, contrast, shot_noise). **Row 3:** selected pairwise combinations. Combined corruptions visually compound the degradation in ways not predictable from individual corruptions.

Table 2: Number of superadditive / additive / subadditive pairs out of 21 total, per model and severity.

Model	Sev.	Super	Add.	Sub
ViT-B/32	1	8	3	10
	3	0	0	21
	5	0	0	21
ViT-B/16	1	6	0	15
	3	0	0	21
	5	0	0	21
ViT-L/14	1	7	2	12
	3	1	0	20
	5	0	0	21

At higher severity ($s \geq 3$), superadditivity collapses to near-zero for all models (table 2). We attribute this to a **floor effect**: at high severity, individual corruptions already drive accuracy near chance (≈ 0.01 – 0.05), leaving no room for further combined degradation. The full degradation picture is shown in fig. 4 (heatmaps) and fig. 5 (severity curves).

4.3 RQ2: Does Scale Help?

table 1 shows that while ViT-L/14 achieves the highest clean accuracy (+8.2 pp over ViT-B/32), the **Gap** between single and combined corruptions is *largest* for the bigger model (0.249 vs. 0.231). ViT-L/14 also exhibits a substantially stronger interaction for contrast + shot_noise (+0.227) than ViT-B/32

Table 3: Top-5 superadditive pairs by interaction effect $\mathcal{I}$ (eq. (4)) at severity $s = 1$. **Exp.** = expected additive drop; **Real** = observed drop.

Model	Pair	Exp.	Real	$\mathcal{I}$
ViT-L/14	contrast+shot_noise	0.543	0.770	+0.227
ViT-B/16	contrast+shot_noise	0.624	0.742	+0.118
ViT-B/32	contrast+shot_noise	0.613	0.724	+0.111
ViT-L/14	fog+shot_noise	0.689	0.799	+0.110
ViT-B/32	fog+contrast	0.330	0.419	+0.089

(+0.111).

Finding 1: Scale does not protect against combined corruptions. Higher clean accuracy creates more headroom for super-additive degradation, amplifying interaction effects.

4.4 RQ3: Which Pairs Are Most Dangerous?

The heatmaps in fig. 4 reveal a consistent ranking across architectures. The most dangerous pairs invariably involve shot_noise combined with a moderate structural corruption (contrast, fog). Pairs of two already-severe corruptions (e.g., gaussian_noise + shot_noise) tend to be *subadditive* due to floor effects. The interaction maps are further visualised in fig. 6. fig. 2 directly compares interaction effects across all three architectures, confirming that ViT-L/14 consistently exhibits stronger superadditivity than smaller models.

Corruption hierarchy. Averaged across all models at $s = 3$, the robustness ranking is: This ranking is **invariant across all**

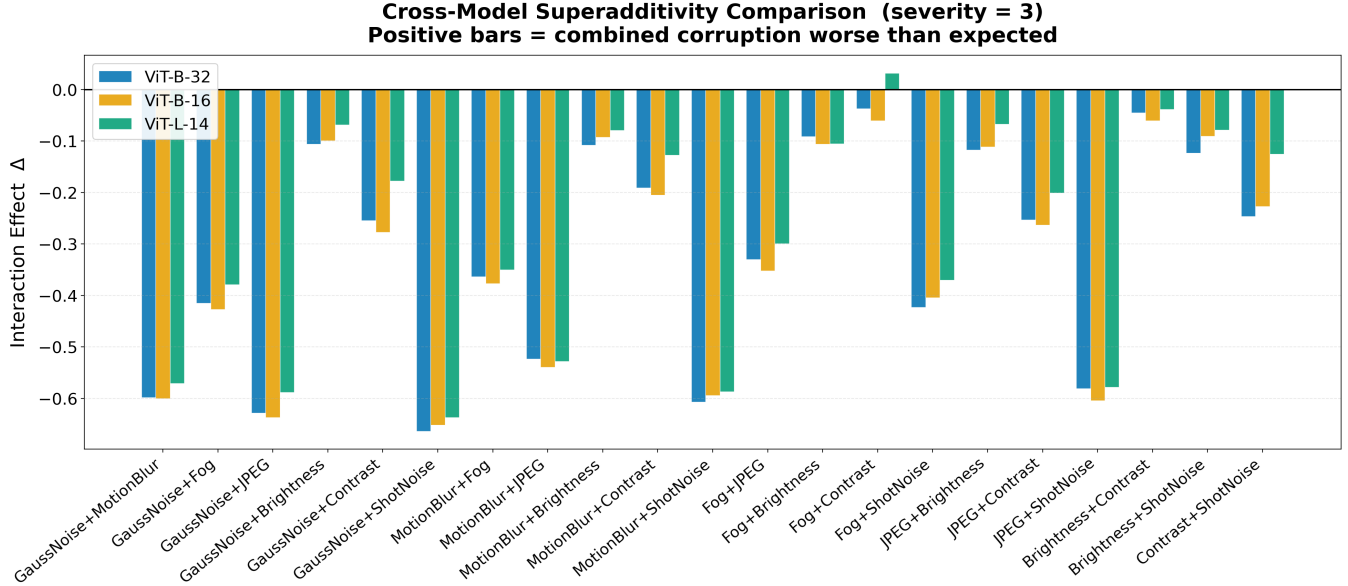


Figure 2: Cross-model comparison of interaction effects $\mathcal{I}$ (eq. (4)) for all 21 corruption pairs at severity $s = 3$. Grouped bars show ViT-B/32 (blue), ViT-B/16 (orange), and ViT-L/14 (green). Positive values indicate superadditive pairs. ViT-L/14 consistently shows the largest positive interactions, confirming that model scale amplifies superadditivity (section 4.3).

Table 4: Single-corruption robustness ranking (averaged across all models, severity $s = 3$). Higher rank = more robust.

Rank	Corruption	Avg. Top-1 ($s=3$)
1	brightness	0.668
2	contrast	0.522
3	fog	0.317
4	jpeg_compression	0.170
5	motion_blur	0.126
6	gaussian_noise	0.097
7	shot_noise	0.089

three architectures, suggesting a structural property of patch-based encoders rather than a parametric one. fig. 3 visualises the robustness profile of each model as a radar chart, making the corruption hierarchy and cross-model differences immediately apparent.

Patch-size trade-off. ViT-B/16 outperforms ViT-B/32 on gaussian_noise at $s = 1$

Patch-size trade-off. ViT-B/16 outperforms ViT-B/32 on gaussian_noise at $s = 1$ (0.341 vs. 0.276, +6.5 pp) but underperforms on jpeg_compression (0.279 vs. 0.290, -1.1 pp). Smaller patches provide finer spatial granularity aiding noise recovery, but JPEG artifacts manifest as high-frequency ringing that is more disruptive to fine-grained patch embeddings.

Finding 2: Patch size creates a noise/JPEG trade-off: ViT-B/16 is more robust to noise but less robust to compression than

Table 5: Order sensitivity of top superadditive pairs on ViT-B/32, severity $s = 1$. $\mathcal{I}$ = interaction effect (eq. (4)). $\Delta_{\text{order}} = |\text{acc}(c_1 \rightarrow c_2) - \text{acc}(c_2 \rightarrow c_1)|$. Large values indicate worst-case evaluation requires testing both orders.

Pair	$c_1 \rightarrow c_2$	$c_2 \rightarrow c_1$	Δ_{order}	$\mathcal{I}_{c_1 \rightarrow c_2}$	Verdict
contrast+shot	0.039	0.214	0.175	+0.107	Sensitive
fog+contrast	0.356	0.146	0.210	+0.052	Sensitive
fog+shot	0.021	0.126	0.105	-0.010	Sensitive
blur+fog	0.122	0.173	0.051	+0.065	Sensitive
blur+contrast	0.259	0.266	0.007	+0.063	Invariant

ViT-B/32.

4.5 RQ4: Does Corruption Order Matter?

table 5 shows that 4 of 5 top pairs exhibit significant order sensitivity ($\Delta_{\text{order}} > 0.01$). The most extreme case is fog+contrast: applying contrast *after* fog yields 0.356 accuracy, while reversing the order collapses it to 0.146—a gap of 0.210.

We hypothesise two mechanisms:

- **Amplification.** When a structural corruption (contrast, fog) is applied first, it removes semantic anchors that would otherwise limit damage from subsequent noise.
- **Partial cancellation.** When noise is applied first, a subsequent contrast adjustment can inadvertently compress the noise range, reducing its perceptual impact.

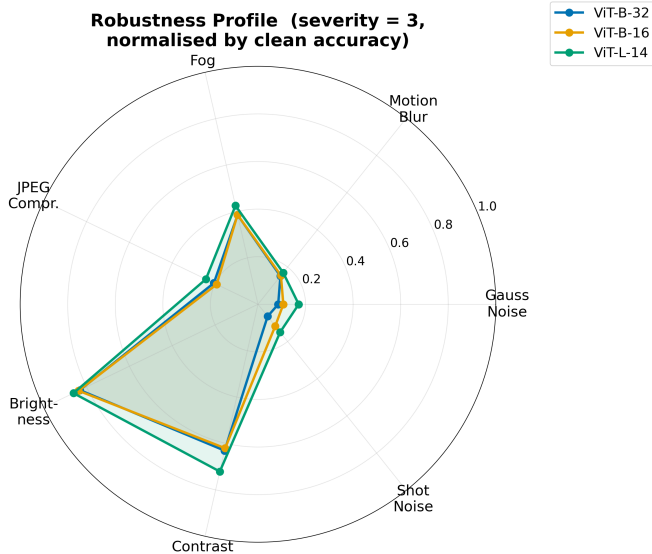


Figure 3: Robustness radar chart for all three models at severity $s = 3$, normalised by clean accuracy. Each axis represents one corruption type; larger enclosed area indicates higher robustness. All models share the same profile shape, confirming that the corruption hierarchy is architecture-invariant. ViT-L/14 (green) occupies the largest area, consistent with its higher clean accuracy.

Finding 3: *Worst-case evaluation must consider both orders of each pair. Fixed-order protocols underestimate worst-case degradation by up to 0.21 in absolute accuracy.*

5 Discussion & Limitations

Why contrast + shot_noise dominates. We hypothesise that these corruptions attack *complementary* feature channels. Reduced contrast removes global luminance structure; shot noise destroys local texture statistics. CLIP’s patch tokenisation relies on both for semantic alignment—losing both simultaneously causes catastrophic failure.

Why scale amplifies superadditivity. ViT-L/14’s higher clean accuracy implies richer, more complex feature representations with more “ways to fail” under compound perturbations. This is an instance of the accuracy–robustness trade-off under compound perturbations.

Limitations.

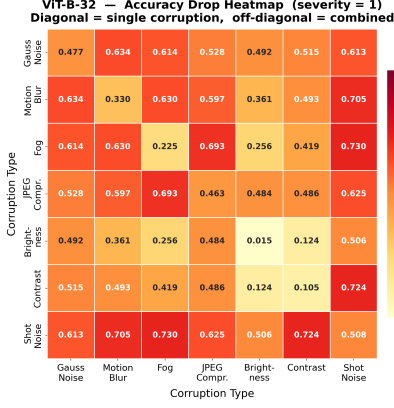
- **Dataset.** CIFAR-100 (32×32 px) is used as an ImageNet proxy due to access constraints; low resolution may exaggerate corruption effects. Replication on full ImageNet is a priority.
- **Corruption scope.** We evaluate 7 of 15+ corruptions in ImageNet-C; extending to all types would yield a more complete interaction map.

- **Pair order.** We evaluate one fixed order per pair in the main study; order sensitivity is analysed in section 4.5 for top-5 pairs only.
- **Beyond pairs.** Triplet and higher-order combinations remain unstudied.
- **No XAI.** Grad-CAM or attention visualisations under combined corruptions would provide mechanistic insight; we leave this for future work.

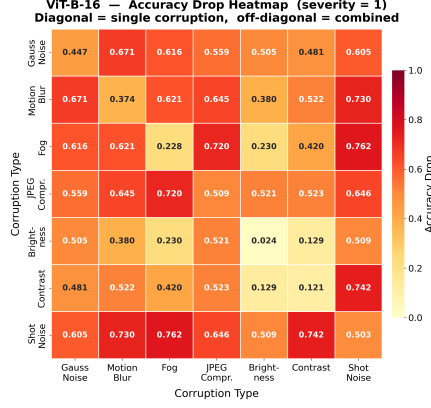
6 Conclusion

We presented the first systematic study of pairwise corruption interactions for CLIP-family visual encoders. Our interaction effect metric $\mathcal{I}$ quantifies whether combined corruptions are worse than the sum of their parts. Superadditivity is a genuine, reproducible phenomenon at low severity, with **contrast + shot_noise** emerging as the most dangerous pair across all three architectures. Model scale *does not* mitigate combined corruption effects—larger models exhibit stronger superadditivity due to higher clean accuracy. Furthermore, corruption order critically determines worst-case degradation, with gaps up to 0.21 in absolute accuracy for fixed-order protocols.

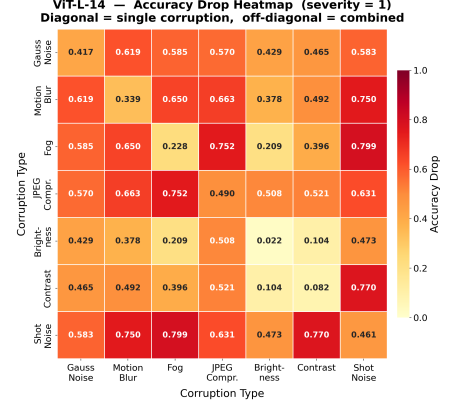
These results expose a systematic blind spot in current VLM evaluation and motivate the adoption of combined-corruption, multi-order benchmarking protocols.



(a) ViT-B/32, $s = 1$

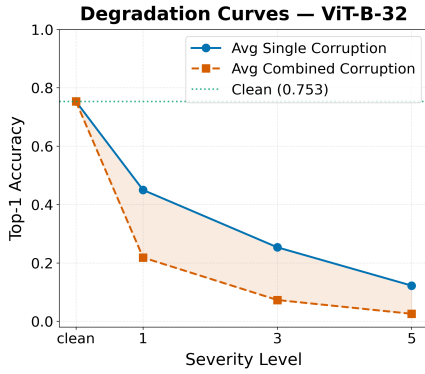


(b) ViT-B/16, $s = 1$

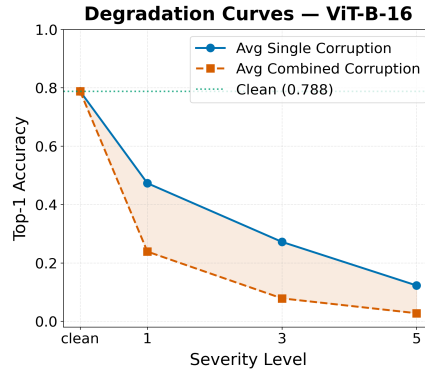


(c) ViT-L/14, $s = 1$

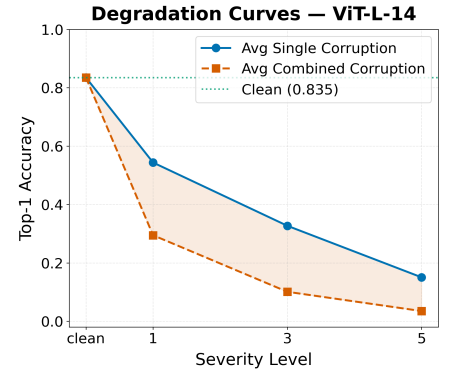
Figure 4: Accuracy drop heatmaps for all pairwise corruption combinations at severity $s = 1$. Diagonal = single-corruption drop; off-diagonal = combined-corruption drop. Darker red indicates more severe degradation. The noise-family quadrant (bottom-right) consistently shows the worst combined performance across all architectures.



(a) ViT-B/32



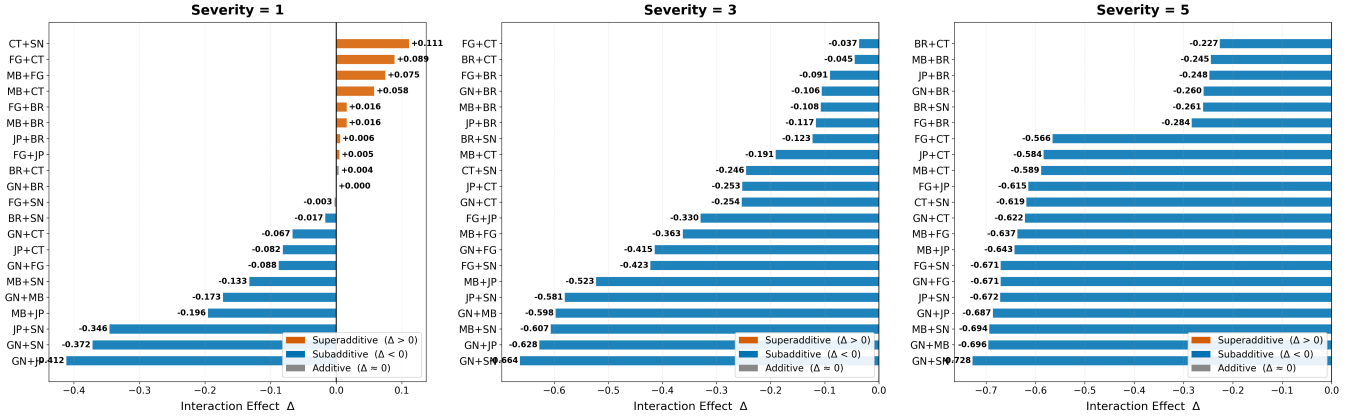
(b) ViT-B/16



(c) ViT-L/14

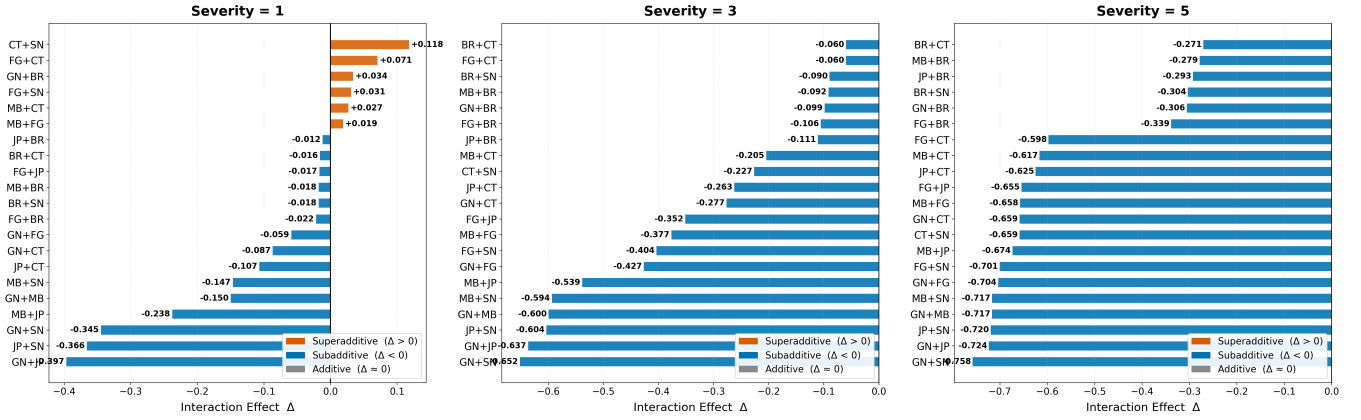
Figure 5: Average Top-1 accuracy vs. severity for single corruptions (blue, solid) and combined corruptions (red, dashed). The gap between curves narrows at high severity, illustrating the floor effect (section 4.2). Clean accuracy is shown as a green dotted baseline.

Superadditivity Analysis — ViT-B-32
Positive Δ = combined corruption worse than sum of parts
(GN=Gaussian Noise, MB=Motion Blur, FG=Fog, JP=JPEG Compression, BR=Brightness, CT=Contrast, SN=Shot Noise)



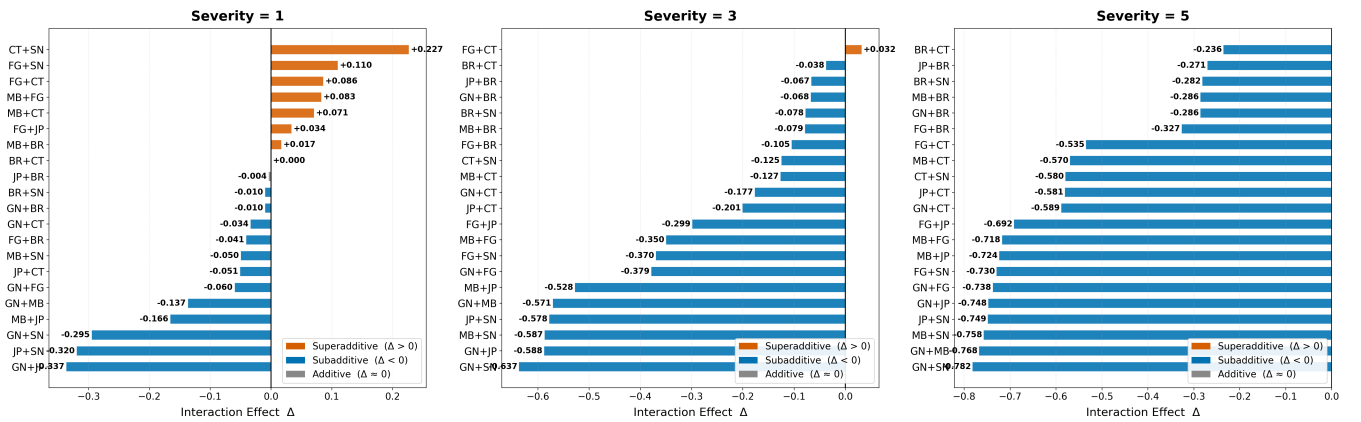
(a) ViT-B/32

Superadditivity Analysis — ViT-B-16
Positive Δ = combined corruption worse than sum of parts
(GN=Gaussian Noise, MB=Motion Blur, FG=Fog, JP=JPEG Compression, BR=Brightness, CT=Contrast, SN=Shot Noise)



(b) ViT-B/16

Superadditivity Analysis — ViT-L-14
Positive Δ = combined corruption worse than sum of parts
(GN=Gaussian Noise, MB=Motion Blur, FG=Fog, JP=JPEG Compression, BR=Brightness, CT=Contrast, SN=Shot Noise)



(c) ViT-L/14

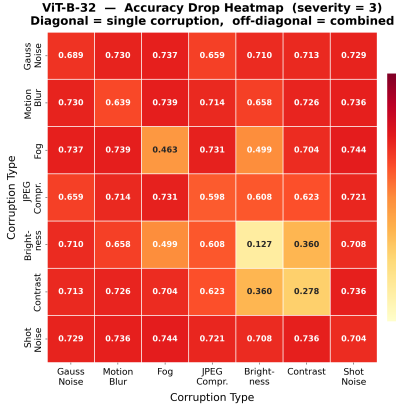
Figure 6: Interaction effect $\mathcal{I}$ (eq. (4)) for all 21 corruption pairs across severity levels. **Red bars:** superadditive ($\mathcal{I} > 0$); **blue bars:** subadditive ($\mathcal{I} < 0$). ViT-L/14 exhibits significantly stronger positive interactions than smaller models. **Abbreviations:** GN=Gaussian Noise, MB=Motion Blur, FG=Fog, JP=JPEG Compression, BR=Brightness, CT=Contrast, SN=Shot Noise.

References

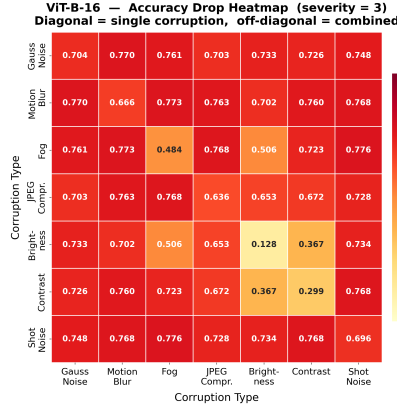
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A Additional Heatmaps

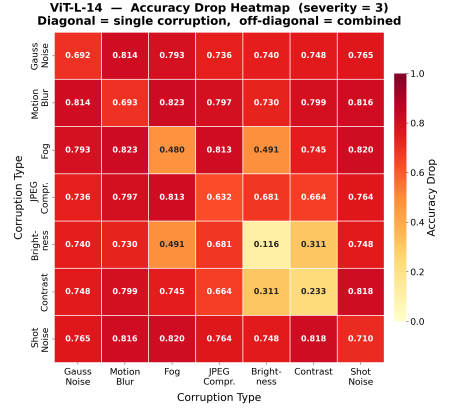
figs. 7 and 8 show accuracy drop heatmaps at severity levels $s = 3$ and $s = 5$. At $s = 3$, the noise-family corruptions have already driven accuracy near zero; at $s = 5$, virtually all pairs exhibit subadditive behaviour due to floor effects.



(a) ViT-B/32, $s = 3$

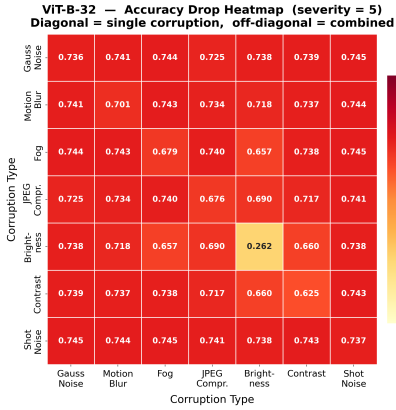


(b) ViT-B/16, $s = 3$

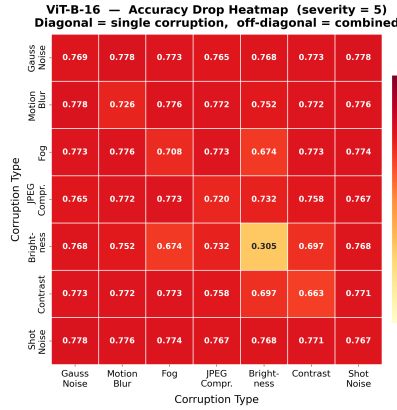


(c) ViT-L/14, $s = 3$

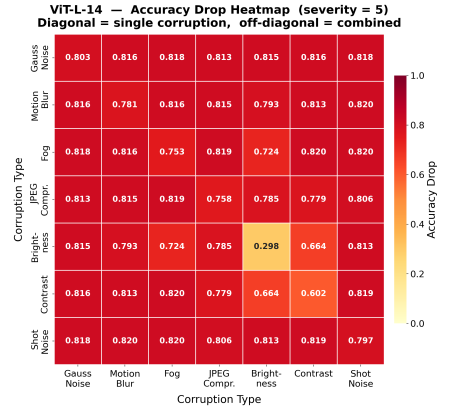
Figure 7: Accuracy drop heatmaps at severity $s = 3$.



(a) ViT-B/32, $s = 5$



(b) ViT-B/16, $s = 5$



(c) ViT-L/14, $s = 5$

Figure 8: Accuracy drop heatmaps at severity $s = 5$.